

# IDIOPATHIC GRANULOMATOUS MASTITIS MIMICKING LOCALLY ADVANCED BREAST MALIGNANCY: A MULTIMODALITY IMAGING CHALLENGE

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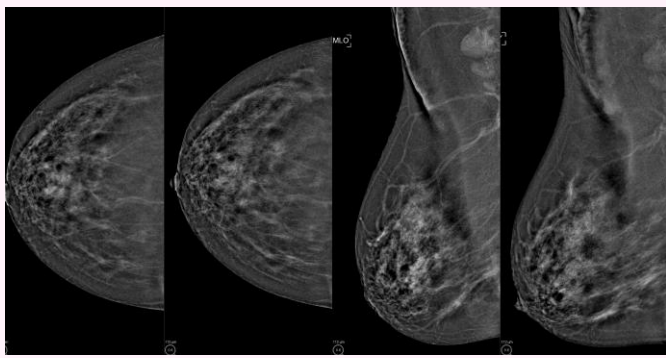
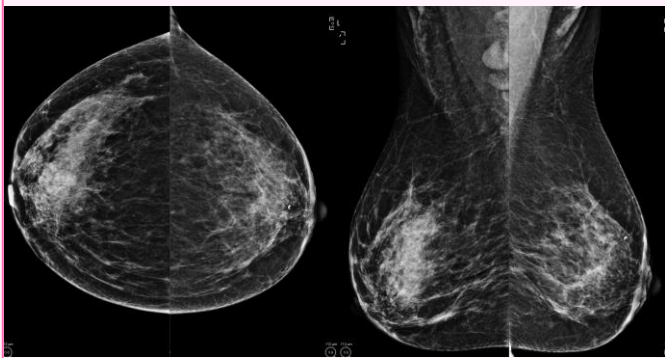
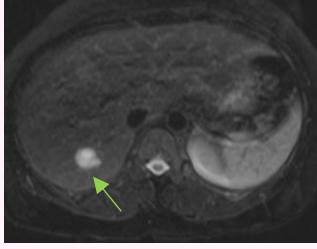
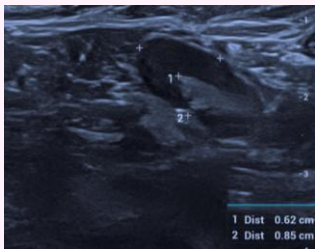
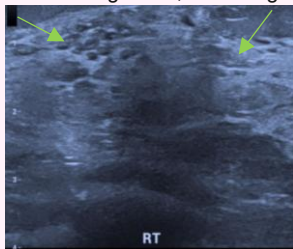


## INTRODUCTION

Idiopathic granulomatous mastitis (IGM) is a benign, chronic inflammatory disease of the breast that predominantly affects women of reproductive age. Despite its benign nature, IGM may closely mimic locally advanced breast carcinoma because of overlapping clinical and imaging features. [1,2] Although the mammographic, ultrasonographic and MRI appearances of IGM are well described, reports detailing its appearance on contrast-enhanced mammography (CEM) remain limited. [4,5] This case highlights the CEM appearance of IGM in a diagnostically challenging presentation, further complicated by enhancing hepatic lesions that initially raised concern for metastatic disease.

## CASE AND MULTIMODALITY IMAGING WORKUP

- ❖ A 42-year-old woman presented with a three-day history of right breast pain and swelling. She has no risk factors for breast cancer.
- ❖ Breast ultrasonography demonstrated irregular heterogeneous hypoechoic area in the right upper outer quadrant and an enlarged ipsilateral axillary lymph node with cortical thickening.
- ❖ Contrast-enhanced mammography (CEM) demonstrated increased density in the right upper outer quadrant on low-energy images, with corresponding regional heterogeneous non-mass enhancement on recombined images. The enhancement decreased on delayed images. An enlarged enhancing right axillary lymph node with cortical thickening was also noted, raising suspicion of malignancy (BI-RADS 4).
- ❖ Abdominal ultrasonography identified, hyperechoic and hypoechoic hepatic lesions. Subsequent CT Thorax Abdomen and Pelvis demonstrated enhancing liver lesions, raising concern for possible metastatic disease. However, liver MRI with Primovist confirmed benign haemangiomas, excluding hepatic metastases.



- ❖ Ultrasound-guided core biopsy revealed non-caseating granulomatous inflammation without evidence of malignancy or specific infection, confirming granulomatous mastitis.
- ❖ Oral corticosteroids resulted in improvement with resolution of pain and inflammatory changes.

## DISCUSSION

- ❖ IGM is a radiological mimic of breast carcinoma, with considerable overlap in sonographic and contrast-enhanced mammographic appearances. Irregular hypoechoic abnormalities, non-mass enhancement and ipsilateral axillary nodal enlargement may closely simulate locally advanced malignancy. [1,2]
- ❖ In this case, CEM demonstrated suspicious non-mass enhancement with reduce enhancement on delayed images, together with cortical thickening of the ipsilateral axillary node. These findings are concerning but remain non-specific, as inflammatory breast disease may demonstrate increased vascularity, enhancement and reactive nodal changes. [2,3]
- ❖ The concurrent presence of enhancing hepatic lesions further strengthened the radiological suspicion of metastatic disease. Dedicated Primovist-enhanced liver MRI ultimately characterised these lesions as benign haemangiomas.
- ❖ Overall, this case highlights the value of multimodality imaging in defining disease extent, resolving indeterminate findings and guiding appropriate tissue sampling, while emphasising that tissue diagnosis and radiology–pathology concordance remain essential to exclude malignancy, specific infection and other inflammatory conditions.

## CONCLUSION

IGM is an important differential diagnosis in women presenting with inflammatory breast symptoms and suspicious imaging findings, as aggressive radiological appearances do not necessarily equate to malignancy. This case demonstrates how suspicious CEM findings and concurrent indeterminate hepatic lesions may compound diagnostic uncertainty, highlighting the importance of multimodality imaging and radiology–pathology concordance.

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